

At erection peaks, where vehicles wait for the crane decides whether its queue reaches the whole yard

Cranes as dock resources, the p curve, the “where to wait” mechanism, the H2 decline and a candidate dock-mouth proposition

Dock-mouth blocking at erection peaks: crane utilisation, waiting position and dock-road granularity

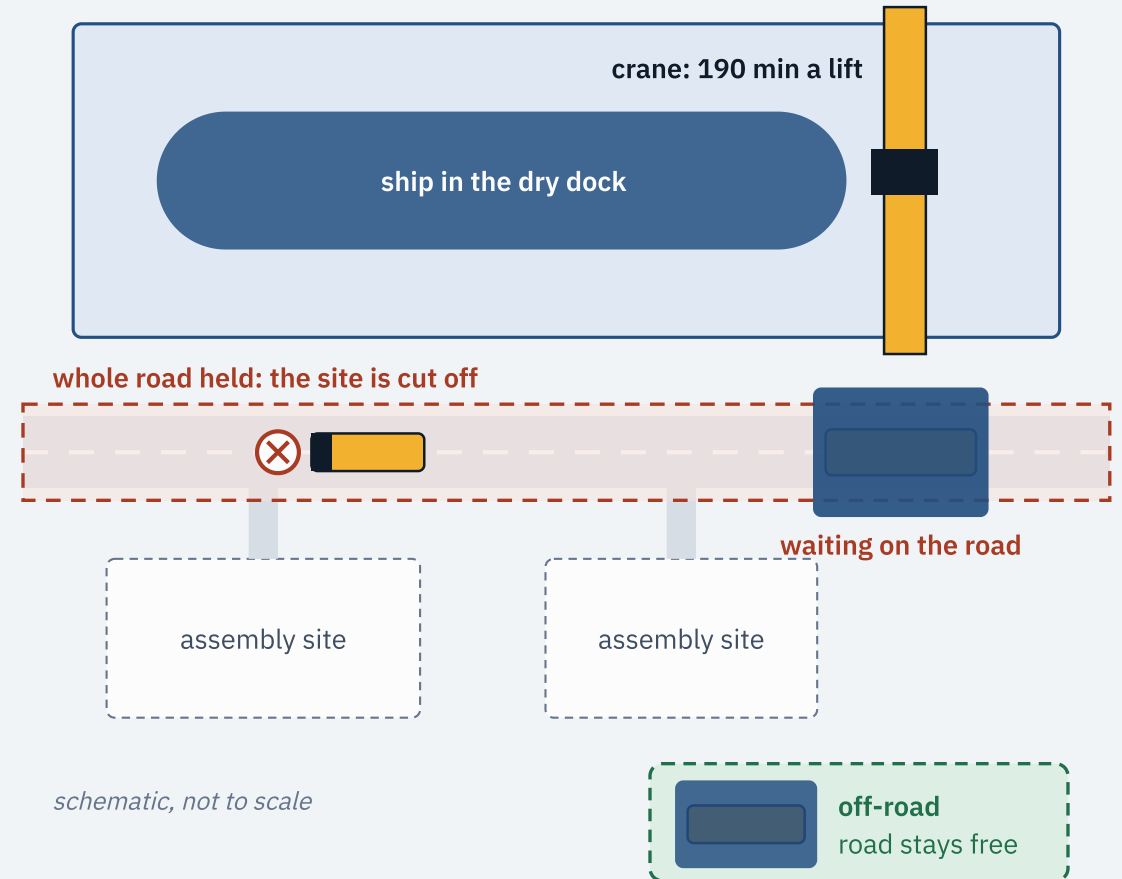
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Basis: thesis-core 83c16bf (T20–T21, reviewed)

Research manifesto v1 · the author's review (9 Oct 2026)

Dock mouth = crane + dock stopping road (in series)



This sub-study in one minute

1

The question

At erection peaks, when does the dock mouth (crane and dock stopping road in series) become the yard's bottleneck? The manifesto makes H1 part 3 conditional: it depends on whether vehicles waiting for the crane block the way to other stops.

2

Readings in hand

Cranes from the dock lists are dock resources; erections are released at crane cadence. On both yards' p curves the plateau of whole-route reservation (waiting off-road) does not move with p ; segment request (reference rule, teleport clearing) falls past $p \approx 0.85$, by 28% with no plateau at Okpo's $p = 0.97$.

3

Mechanism and modelling

The difference is where vehicles wait. So far the decline appears only with teleport clearing and road 161 as one resource; in the author's probe, splitting the road at its stops cut the daily decline at $p = 0.97$ from about -20% to about -5% . After the imagery check, the segmented dock road is Okpo's main scenario from T22 on.

4

Next

T22: safe release without teleporting and a no-clearing switch, a segmented dock road, the crane starvation rate. T25: a three-resource network for the threshold p^* . T26: floating docks and a finite holding area, $K_0(S)$. If the contrast holds under physical rules, the dock-mouth proposition follows.

The sentence to test: whether a crane queue reaches the whole yard depends on where the vehicles queue.

The order of this briefing (click to jump)

- 1

Question and position
 Manifesto items served: H1 part 3, H2, the candidate proposition, H7

p. 4
- 2

Cranes and the p curve
 Crane lists, erection cadence, working points; the p curve under two rules

p. 5
- 3

Mechanism and modelling choices
 Where vehicles wait; teleport clearing and the whole-road resource; the author's probe

p. 7
- 4

When the decline appears (H2)
 Decline readings at the dock; three switches; the T4 small network

p. 10
- 5

The proposition and floating docks
 Starvation caused by blocking; the three-resource network; floating docks and H7

p. 12
- 6

Progress, plan and journals
 Established and in progress; tasks and outputs; candidate journals

p. 14

Question and position: the dock-mouth section of chapter 3, serving H1 part 3, H2, the candidate proposition and H7's floating-dock case

At erection peaks, through what channel does the crane queue become a capacity loss for the whole yard, and can the waiting position and the granularity of the dock road keep it at the dock mouth?

MANIFESTO ITEM	TEXT (RESEARCH MANIFESTO V1)	THIS STUDY SUPPLIES
H1, part 3	Whether the dock mouth becomes the yard's bottleneck depends on whether vehicles waiting for the crane block the way to other stops, that is, where they wait and whether the dock road is used in sections.	ρ curves $\times$ two waiting positions $\times$ two road granularities
H2	As long as the extra vehicles occupy network resources, throughput falls beyond K_0 , and K_0 moves right with the holding-area capacity S .	Decline readings at the dock; a sweep of three switches; T4
Candidate	Dock-mouth proposition: with nominal load $\rho < 1$, starvation caused by blocking destabilises the system; with off-road waiting or a dock road used in sections, stability is restored.	The three-resource network and the crane starvation rate
H7 case	Multi-transporter carriage of giant blocks for floating docks.	A floating-dock scenario (quay-side sink + floating crane)

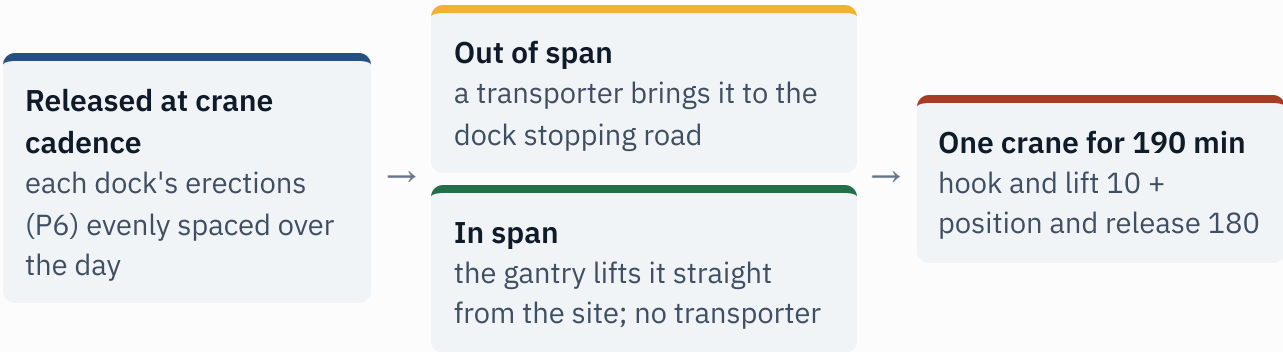
Related claims: C2's falling branch appears “when waiting crowds the road”; C3's orchestration includes waiting points and block granularity. Each conclusion states its conditions once: traffic rule, task-volume basis, resource granularity, yard and map version (manifesto §5).

Sources: research manifesto v1 §2 (C2, C3), §3 (H1, H2, H7), §4 (T4 and the candidate proposition), §5 (wording rules). The English and Korean of manifesto items are faithful translations; the Chinese governs.

Cranes are dock resources: erections are released at crane cadence, and in-span lifts take the crane but not the road

YARD · DOCK	CRANES COUNTED (CAPACITY ≥ 100 T)	NO.
Okpo · dock 1	900 t gantry, 200 t jib crane	2
Okpo · dock 2	900 t gantry	1
Yantai · dock 1	20,000 t bridge crane (Taishan), 370 t gantry	2
Yantai · dock 2	370 t gantry	1

Jib and tower cranes below 100 t (the mean block weight) are not counted; assigning Yantai's two 370 t gantries one per dock is a scenario choice. Okpo gantry spans: 206 m on dock 1 (Riviera 2014), 148 m on dock 2 (The Motorship).



$$\rho = P6/\text{day} \times 190 \text{ min} \div (n_{\text{crane}} \times 960 \text{ min})$$

Three cranes erect at most $3 \times 960 \div 190 = 15.2$ a day

0.88 (0.78-0.97)
Okpo: $P6 = 2 \times 20 \text{ ships} \times 100 \div 300 = 13.3$ a day (range 11.9–14.7).
20 = LNG carriers delivered in 2025 (estimated from IR), 100 = midpoint of DSME's patent “usually 90–110”, 2 = peak day. **A scenario reading, not part of H1** (author, 9 Oct 2026).

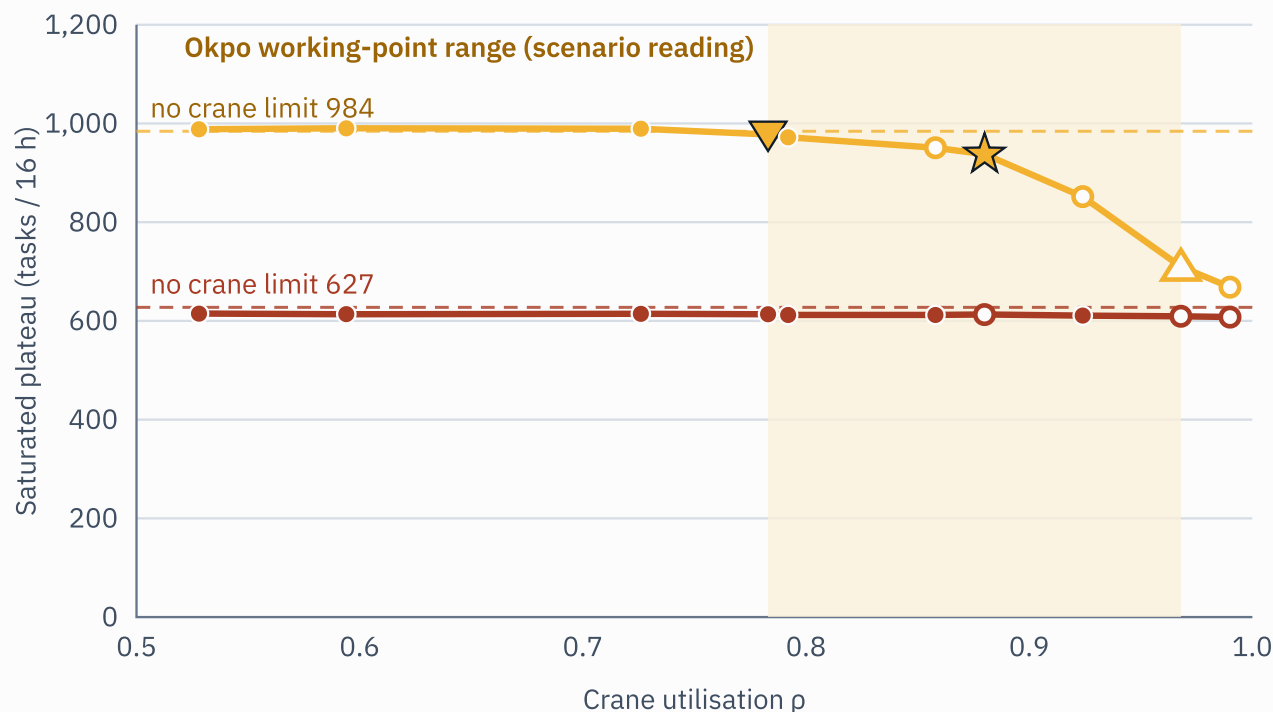
0.13
Yantai: $P6 = 2$ a day from public output; the cranes are far from tight.

12.5%
Okpo's in-span share: zone 018 → dock 1 and zone 014 → dock 2 (span cover ≥ 0.5 on the map); none at Yantai.

Sources: thesis-core 83c16bf, T20 report §1 (crane lists: Hanwha Ocean Facilities page; CIMC Raffles 2012 EIA notice; Hoist Magazine 2008), T21 report §2–3, t21_span.csv, t21_p6.csv. Crane cycle from ZhangSG2025, lines 2525–2528 (another yard); units per ship from DSME patent KR20090050364A and Roh & Lee 2006, p. 116 (89); ships a year estimated from Hanwha Ocean IR.

The ρ curve: the reservation plateau does not move with ρ ; segment request (reference rule, teleport clearing) starts to fall past about 0.85

Okpo Yantai



608-615

Reservation plateau (waiting off-road) over every ρ ; 627 with no crane limit

$\leq \pm 1.2\%$

Segment request (reference rule) for $\rho \leq 0.79$, against 984 with no crane limit

-4.7% / -27.9%

$\rho = 0.88$ (main value, steady) / 0.97 (upper end, falling for 8 days, no plateau)

1.46-1.47

Dock margin of the LP with P6 fixed, nearly constant over ρ : the bound cannot see queue-induced time blocking

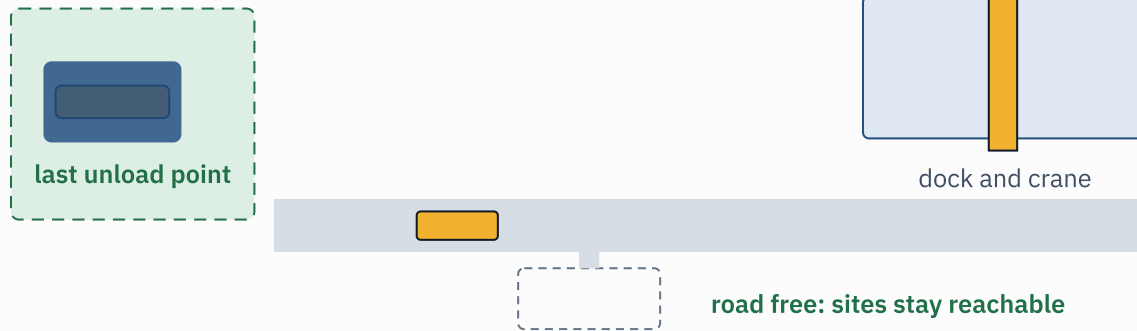
■ Whole-route reservation ■ Segment request · reference rule (teleport clearing)

○ hollow = not steady

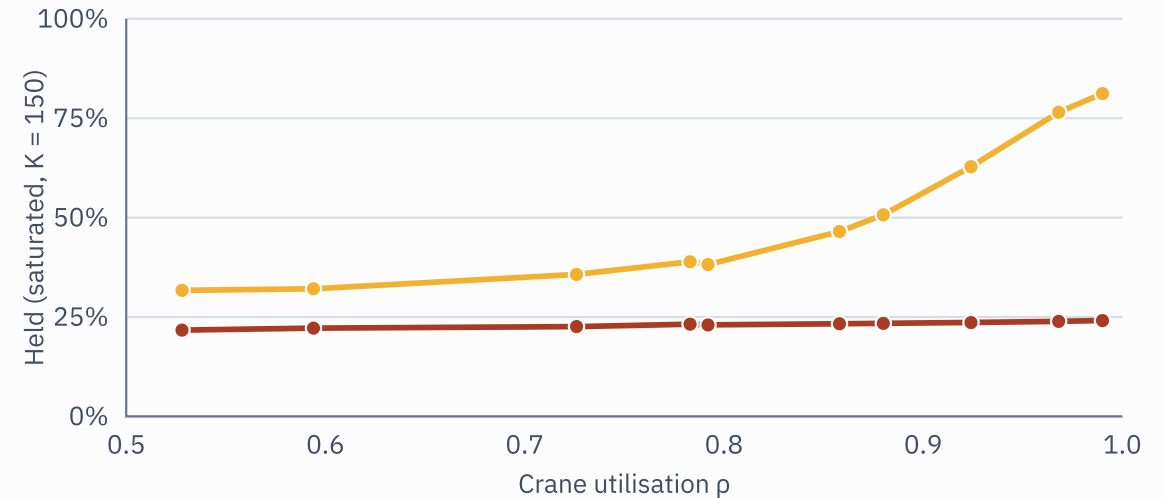
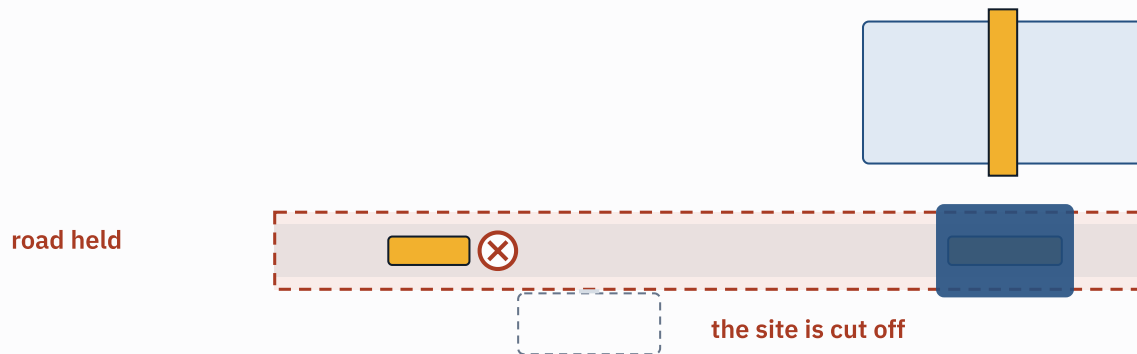
Sources: thesis-core 83c16bf, results/t21_report_9eb8bf099d3a/t21_rho.csv and ρ batches 1b06c6d46f0d (Okpo), 87d02d32bc13 (Yantai). Conditions: Okpo derived map rev 1825, Yantai r396; MY-B flow mix (Roh & Cha 2011; daily volume a 4-transporter subset, Yim 2008, against 300-600 a day yard-wide, Yim 2008 and Shen et al. 2018, rerun in T22), P6 at crane cadence with in-span lifts; saturated demand, 8 days, plateau = mean over $K = 100-150$, 10 seeds; road 161 as one resource; segment request is the reference rule (teleport clearing after 900 s). Hollow: at least one K in 100-150 fails the steady-state test (mix shift $\leq 2\%$, daily trend $\leq 5\%$).

The mechanism: the difference is where vehicles wait – at their last unloading point (off-road) or on the dock road

Reservation: crane slot booked with the route



Segment request (reference rule): ask at the dock



Whole-route reservation Segment request (reference rule)

- The cost moves off-road:** at $p = 0.88$ and $K = 150$, reservation vehicles wait 820 min a day for the crane off-road (segment request 430 min), and the dock stopping road is held only about 23% of the time
- In-span lifts** (same $p = 0.86$, 12.5% of P6 without a transporter): waiting for the crane -29% / -25% , dock road held about -10% , plateau unchanged (951 vs 952). p sets the strength; the in-span share only changes how many vehicles queue on the road

The decline currently rests on two modelling choices: teleport clearing of deadlocks, and the 1.1 km road 161 as one resource

① Teleport clearing: deadlocks a day under segment request (Okpo main scenario)

26

regular day (K = 40)

83

peak day (K = 150)

1,911

saturated, K = 100

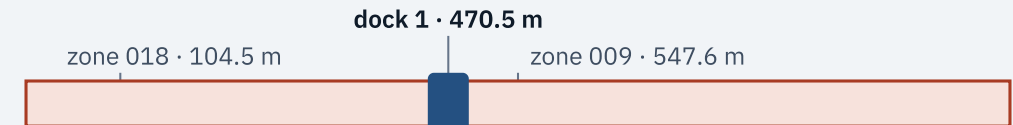
2,657

saturated, K = 150

Each deadlock removes at least one vehicle from the network for 900 s, releasing all it holds. At K = 100 at least 478 vehicle-hours a day, 30% of the fleet's day, are off the network.

Segment request's plateau of 984 and the decline on the p curve are both read under this rule; it is a reference only and cannot support a C or H conclusion on its own (manifesto §5, rule 2).

② Road 161: 1,095 m long, 9 m wide, three stops along it



One resource (current): one vehicle at the dock holds all 1,095 m



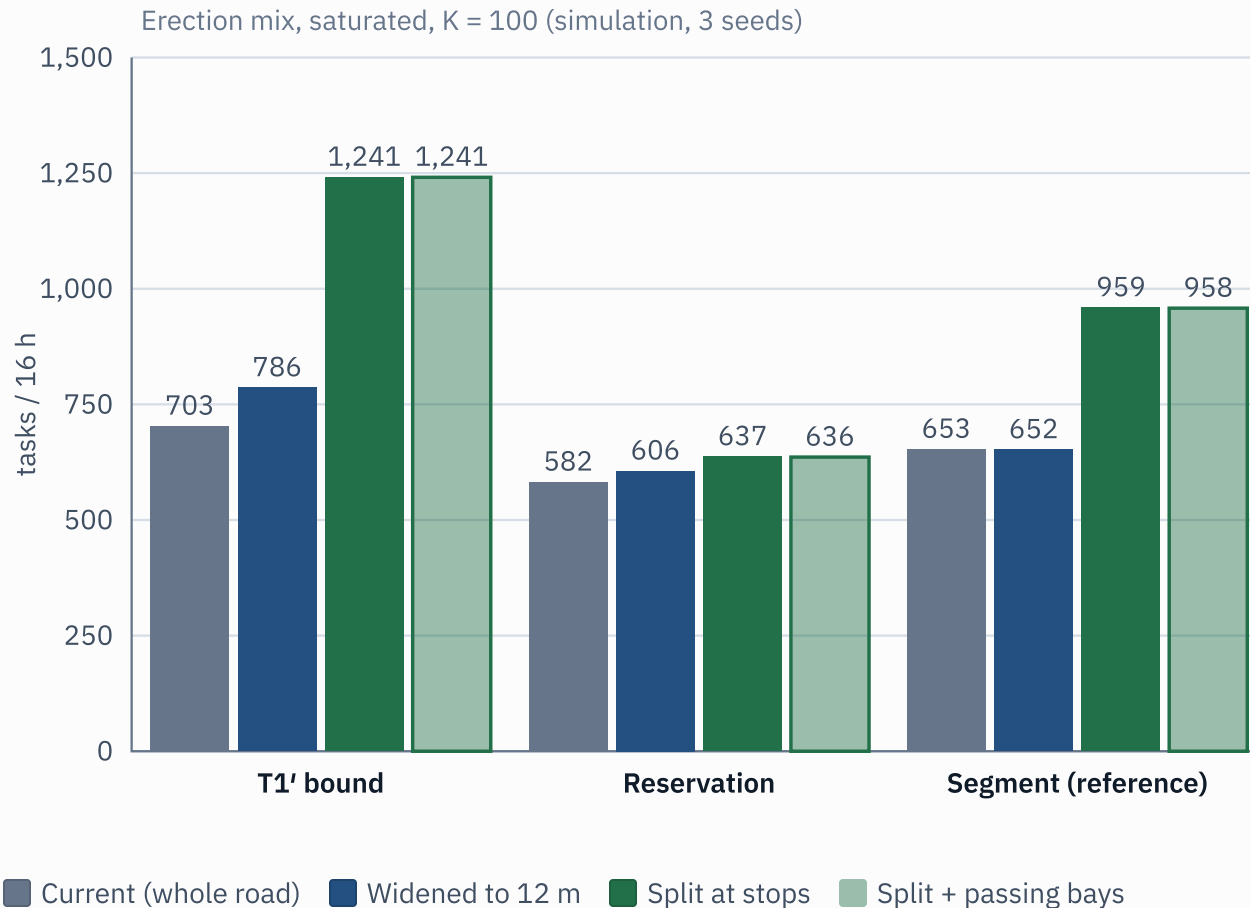
Split at stops: bounds 25 m either side, 7 sections; only the 50 m at the dock is held

0 m 1,095 m

Sectioning is borrowed from railway block sections (UIC 2013); single-lane mine roadways are run on block-section logic too (Eustace et al. 2021).

The author's imagery check (9 Oct 2026): while a vehicle unloads or waits at the dock edge, the neighbouring assembly sites **stay reachable**. From T22 on, the road split at its stops (section = "stops") is Okpo's main scenario; the whole road is the comparison.

The author's probe: when the dock mouth dominates, splitting the dock road at its stops does more than widening it (3 seeds, a prototype)



- **Split at stops:** T1' 703 → 1,241 (+77%); at K = 100, reservation +10%, segment request (reference rule) +47%
- **Widened to 12 m:** T1' +12%, reservation +4%, segment request 0; passing bays on top of the split add nothing
- **p = 0.97, K = 120:** segment request's daily trend -21%, -44%, -1% → -7%, -2%, -7% (about -20% → -5%, at the edge of the ±5% tolerance); reservation falls under no variant

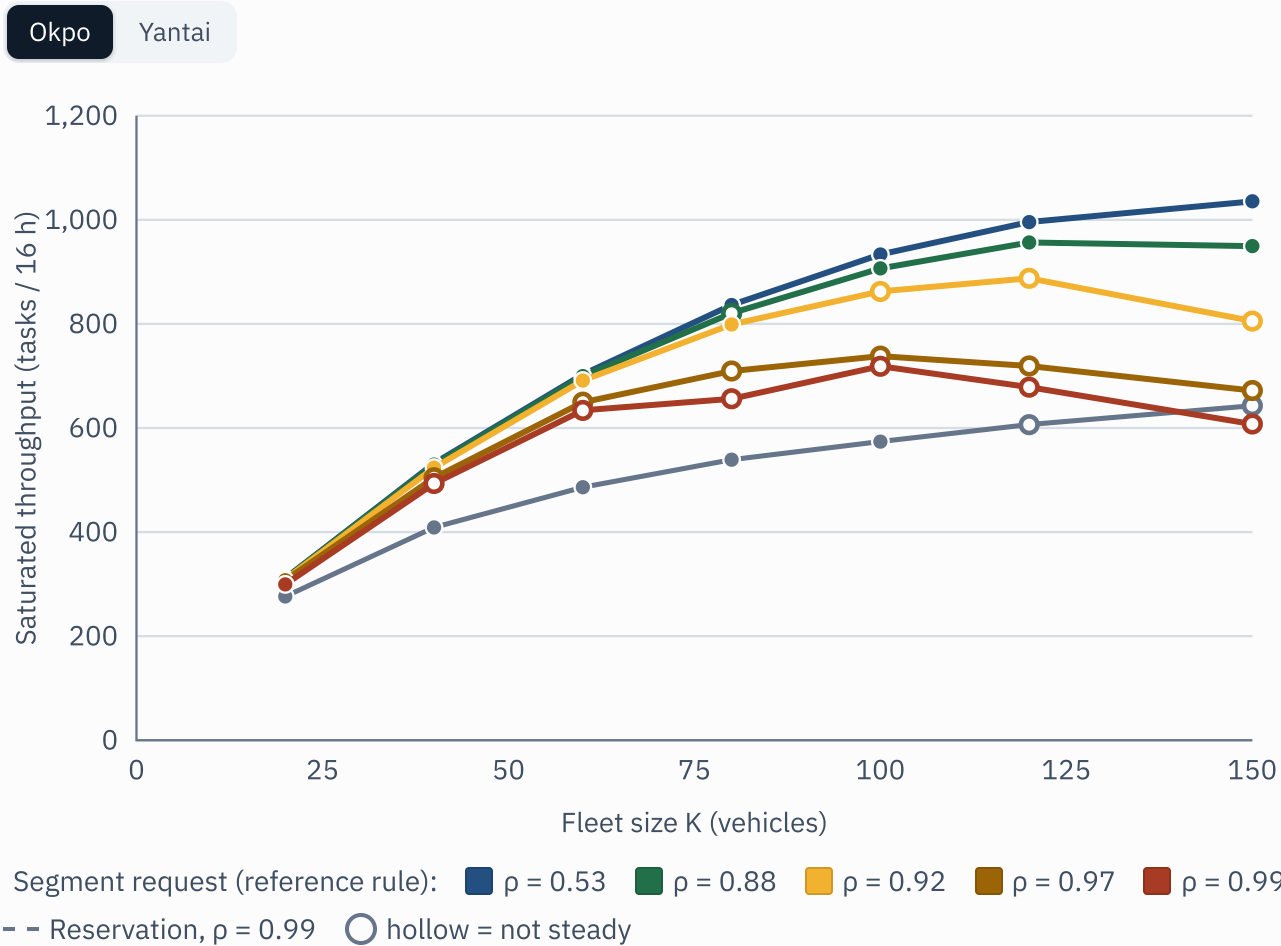
Two confounders (from the review)

① The split had no direction locks, so deadlocks rose (1,752 → 2,329 a day at $p = 0.97$) and teleporting feeds segment request's gain; only reservation's change counts at face value. ② Why the crane idles is unverified: the vehicle at the dock edge may have been teleported away.

Next: T22 reruns the split with direction locks and without teleporting, and checks whether teleported vehicles stood at the dock edge; T25 makes the first upgrade table (10 seeds, full K grid).

Source: the author's review (9 Oct 2026), appendix A; numbers read from its tables by make_data.py. Code of thesis-core 83c16bf unchanged, Okpo derived map rev 1825; bounds over 10 seeds, simulation over 3 seeds (101–103). Erection mix: P6 15%, saturated, K = 100, 4 days; crane cadence: $p = 0.97$, K = 120, 8 days. Variants: widened = road 161 from 9 m to 12 m; split = bounds 25 m either side of the 3 stops, no direction locks; passing bays = a 60 m unrestricted section in the middle of each of the two long gaps after the split.

H2, when the decline appears: while vehicles waiting for the crane hold the road, saturated throughput first rises and then falls with fleet size



With off-road waiting and a deadlock-free rule, throughput never falls as vehicles are added: a plateau. As long as the extra vehicles occupy network resources, throughput falls beyond K_0 , and K_0 moves right with the holding-area capacity S .

YARD	CRANE LOAD	THROUGHPUT AS K GROWS (SEGMENT, REFERENCE RULE)
Okpo	$\rho = 0.97$	738 → 719 → 672 ($K = 100 / 120 / 150$)
Okpo	$\rho = 0.99$	719 → 608 ($K = 100 \rightarrow 150$)
Yantai	$\rho = 0.92$	480 → 291 ($K = 40 \rightarrow 150$)
Yantai	$\rho = 0.99$	349 → 209 ($K = 40 \rightarrow 150$)

Whole-route reservation (off-road waiting, deadlock-free): on both yards and at every ρ , throughput rises monotonically with K ($K = 20\text{--}150$).

The decline readings are non-steady and come from segment request (reference rule, teleport clearing), and Okpo's also from road 161 as one resource; they do not support H2 on their own, but mark where to sweep under rules without teleporting (T22, T26).

Sources: research manifesto v1 §3, H2 (quoted); thesis-core 83c16bf ρ batches 1b06c6d46f0d and 87d02d32bc13 (saturated, 8 days, mean of 10 seeds); the table matches the author's review §2.5 (checked item by item by make_data.py). Other conditions as on the ρ -curve slide. Hollow: that K fails the steady-state test.

Testing H2: sweep three switches one at a time under rules without teleporting; prove T4 first on a single-lane corridor with a holding area

① Holding area S

idle and waiting vehicles beyond S occupy the roadside T26

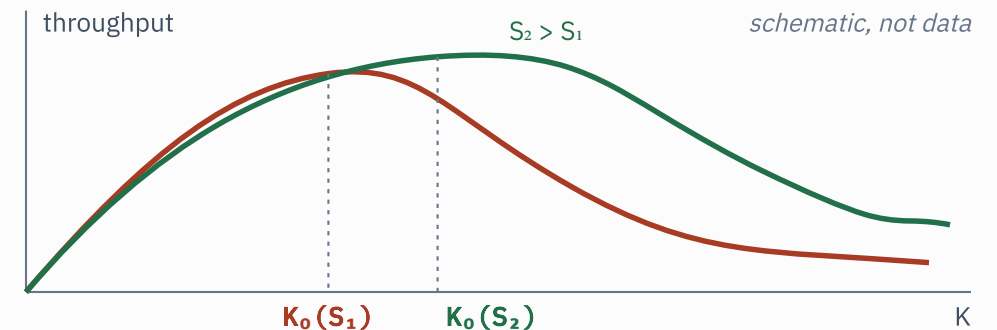
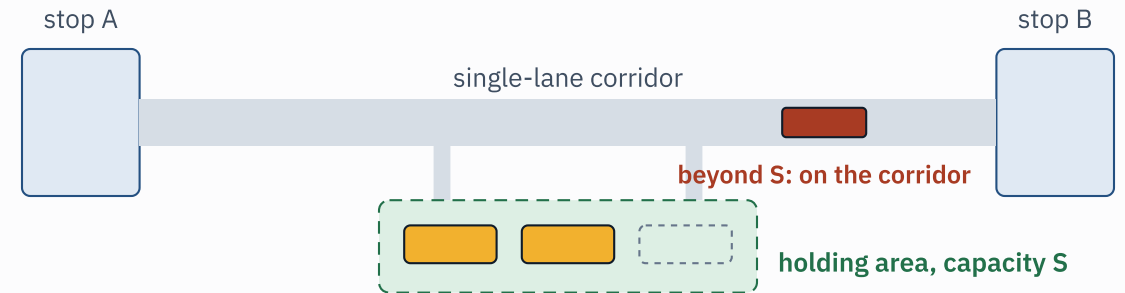
② Clearing mode

safe release segment_safe (Banker-type deadlock avoidance, no vehicle leaves the network, as signalmen release vehicles on site, KED Global 2024); no clearing: stuck vehicles keep the road T22

③ Where to wait for the crane

off-road (booking to the dock, a holding bay at the dock) or on the dock road; the road whole or split at its stops T22 · T25

If it holds: the conditions under which more vehicles hurt, and “how many vehicles a holding area is worth”. **If not:** a redundancy index, the ratio of marginal throughput to the free-flow margin. Good orchestration can push K_0 back.



T4, non-monotonicity (manifesto §4): on a single-lane corridor with a holding area of capacity S , under greedy release, throughput falls strictly for $K > K_0(S)$. Method: a sample-path construction. Output: chapter 5's design rule “how many vehicles a holding area is worth”.

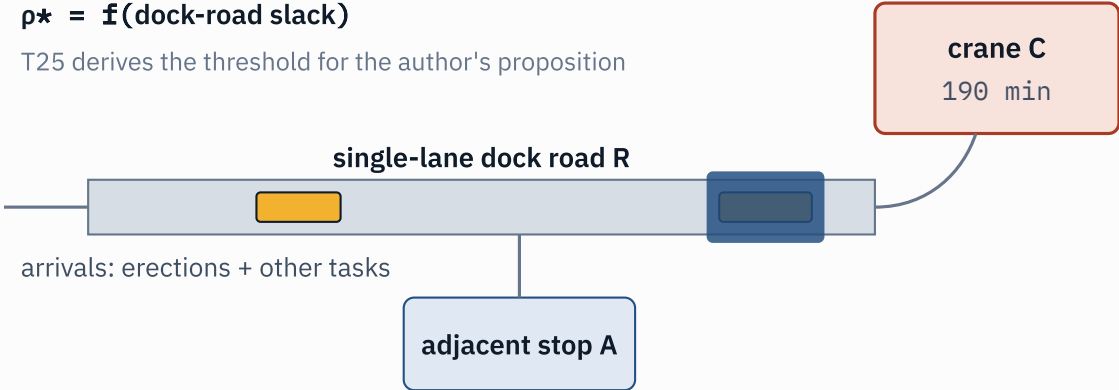
The candidate dock-mouth proposition: with nominal load $\rho < 1$, starvation caused by blocking can still destabilise the system

- **The channel:** vehicles waiting for the crane block the way to other stops
→ later erections cannot reach the dock and the crane idles because upstream is blocked → effective capacity falls below 1, and the system is unstable although nominal $\rho < 1$
- **Restoring stability:** waiting off-road (reservation, a holding bay at the dock) or a dock road used in sections → the stable range returns to $\rho < 1$
- **Observable:** the crane starvation rate, the share of time a crane is free while a released erection has not started, with the part where the vehicle is stuck in the network reported separately (T22)

THEORY SOURCE	WORKS	USE
Queueing networks with blocking	Onvural 1990; Balsamo et al. 2001 to verify	foundation
Policy-induced instability at nominal load < 1	Kumar & Seidman 1990; Lu & Kumar 1991; Rybko & Stolyar 1992 from memory, to verify	analogy, transfer source
Fluid models	Dai 1995 to verify	proof tool

$\rho^* = f(\text{dock-road slack})$

T25 derives the threshold for the author's proposition



Wording: we extend the blocking-induced starvation of queueing networks with blocking to a dock mouth where a crane is in series with an exclusively occupied road. Status: a candidate in the manifesto, to be decided on T22 item 2.

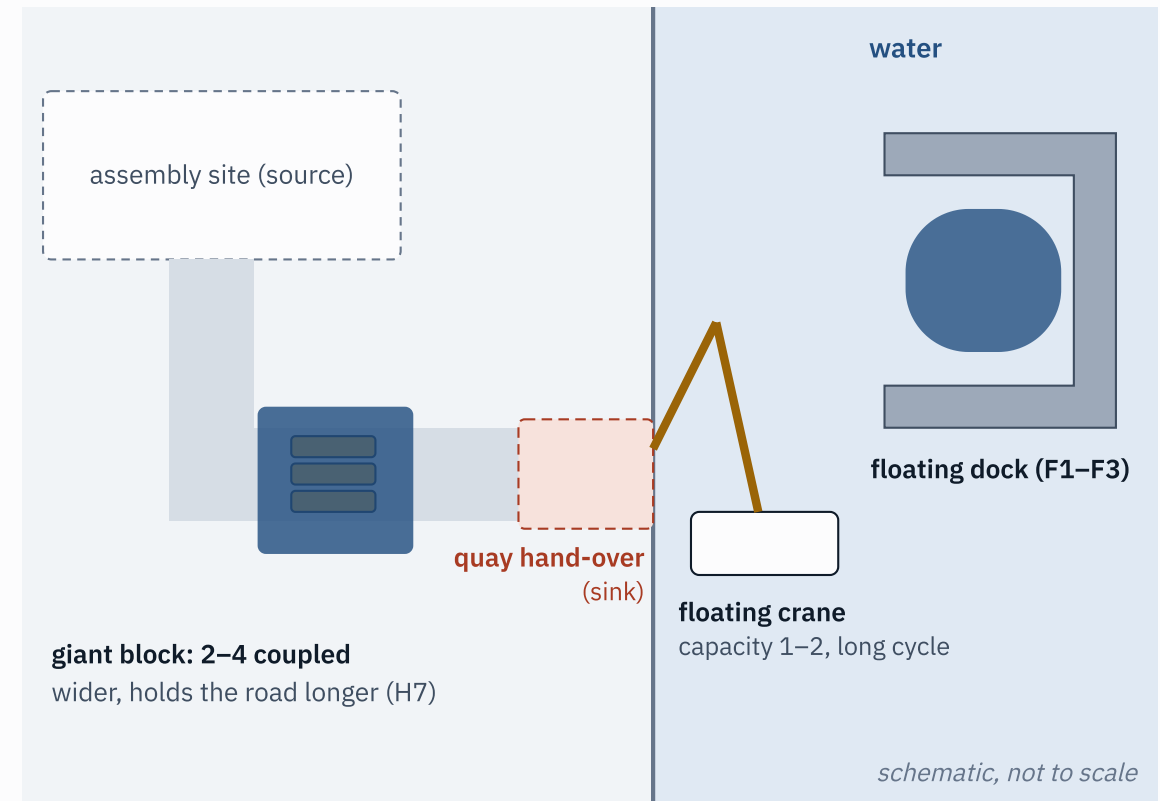
What this adds over the closest work in the field: ZhangSG2025 solves gantry-crane scheduling and transporter scheduling as two separate problems; here crane cadence and road-owning loaded vehicles sit in one simulation and one bound (the crane row of T1'), which reveals the bottleneck of the two in series. Whether the field already has a joint occupancy model is being checked by the literature library (T24).

Sources: research manifesto v1 §4 (candidate); the author's review §2.2, strengthening item 2, and appendix B (transfer sources and theory, evidence levels as given); runs/2026-10-09_T22_prompt.md §2.4 (crane starvation rate); T20 report §7 and T21 report §7 (comparison with ZhangSG2025).

Floating docks: a quay-side sink plus a floating crane; giant blocks on coupled transporters are H7's case

- **Three floating docks:** Hanwha Ocean 4Q24 IR, p. 29, “Floating Dock : 3EA”; on the imagery (rev 1833) F1 $\approx 457 \times 82$ m, F2 $\approx 360 \times 77$ m, F3 $\approx 472 \times 80$ m, with container ships being built in F1 and F3
- **Fits the base scenario:** LNG carriers in dry docks 1 and 2 (confirmed by the author; dock 1 builds four at once, Seoul Shinmun 2024, Hanwha 2025); all 33 merchant ships there would give $p = 1.29\text{--}1.60$, beyond the cranes
- **Model:** source = assembly sites near the quay; sink = the hand-over stop at the quay edge; floating crane = capacity 1–2 and a long cycle, reusing the crane code
- **Giant blocks:** 2–4 coupled transporters, wider and holding the road longer, returning empty as a team $\rightarrow$ H7, linking to MY-B's coupled carriage

The practice is general knowledge; each item needs a public source before modelling (T24), including “about 10 giant blocks a ship”. The base scenario stays; the floating-dock flow is separate (T26).



Established and in progress: each next task feeds a thesis figure or section

ITEM	STATUS	OUTPUT (WHERE IN THE THESIS)
Cranes as dock resources, erections at crane cadence, in-span lifts; the crane row and window edge term of T1' (none of 17,760 occupancy points on the two yards exceeds it)	established · T20–T21	Chapter 3 methods and parameter table
Both yards' p curves: reservation and segment request (reference rule), whole road	established · comparison	Comparison curves in the chapter 3 dock section
The author's probe: split, widening, passing bays (3 seeds)	prototype	Design of T22 and T25
T22: safe release segment_safe and a no-clearing switch; dock road split at its stops (direction locks for the reference rule); crane starvation rate; p curves and erection mix redrawn on both yards	running	The mechanism figure of the chapter 3 dock section; readings for H1 part 3
T25: the three-resource dock network (p^* against dock-road slack); upgrade simulations: split, holding bay, passing bays, widening (10 seeds, full K grid); capacity against block length	queued	The proposition's threshold; the first road–vehicle exchange-rate table (chapter 5)
T26: Okpo rev 1833 and the floating-dock scenario; finite holding area S and $K_0(S)$	queued	H2's $K_0(S)$ curve and “how many vehicles a holding area is worth”; H7's floating-dock case
T24: transfer sources on queueing networks, policy-induced instability and block granularity; public sources on floating docks and giant blocks	running	Reference checks; floating-dock parameters

Risk

Under physical rules and the split road, Okpo's dock-mouth decline weakens or disappears.

Mitigation

Write the result as conditional readings on where to wait and how to split, plus design levers, not as one yard's threshold; yards in chapter 6 with narrower or longer dock roads add cases.

Sources: 00_总控/tasks.md T20–T26 (9 Oct 2026); reviews/T20.md, T21.md; T21 report §5 and §10 (case-by-case check of 17,760 points); the author's review §4.1 and §5 (risks and mitigation). T22–T26 have no results yet; they are listed as plans.

Candidate journals: options now, a choice once the proposition and the physical-rule results are in

Theory (if the proposition is proved)

Operations Research
Stochastic Systems
Queueing Systems

Selling point: blocking-induced instability, an explicit threshold p^* , and stabilisation by off-road waiting or a sectioned road.

Transportation and OR

Transportation Science
Transportation Research Part B

Selling point: the capacity interval, the dock-mouth mechanism and proposition, as a section of the first paper (chapter 3) or a paper of its own.

Applied

IISE Transactions
Transportation Research Part E
European Journal of Operational Research

Selling point: design rules on waiting position, road sectioning and floating docks, and upgrades priced in vehicles.

The author's review §4.2 on the first paper: submit to TR-E or IISE Transactions; with the attainability of T1' and a proof of the proposition, aim at TR-B or Transportation Science, adding “and dock-mouth instability” to the title. All are options for the author to choose later.

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To close: the dock-mouth section answers “where to wait, and how to split the road”

1

The crane is in the model: erections at crane cadence, in- and out-of-span from the map, and one parameter p places a yard on the curve.

2

The mechanism shows: with off-road waiting the plateau does not move with p ; when vehicles hold the dock road while waiting (reference rule, whole road), the crane queue reaches the yard.

3

Next, quantify it under physical rules and a split road, add the threshold p^* of the three-resource network, and turn it into chapter 3's dock section and chapter 5's design rules.

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Thank you.